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Relationships between the erythrocyte sedimentation rate, plasma proteins and viscosity, and leucocyte counts in thoroughbred racehorses

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The influence of plasma proteins on erythrocyte aggregation was studied in a population of young thoroughbred racehorses, using the 60 minute erythrocyte sedimentation rate (ESR) with and without haematocrit standardisation. The ESR was correlated inversely with the haematocrit, but directly with fibrinogen, plasma viscosity and serum total globulins. When ESR values were standardised to a common haematocrit the correlation coefficients for the same plasma protein factors were increased. Albumin levels showed a strong direct relationship with haematocrit which accounted for the inverse correlation found between albumin and ESR. The haematocrit standardised ESR showed no significant correlation with albumin levels. Total leucocyte and absolute neutrophil counts were not correlated with either ESR or haematocrit standardised ESR. The high correlation ($r = 0.75$) found between fibrinogen levels and haematocrit standardised ESR suggests that differences in this acute phase protein influence the degree of red cell aggregation and rouleaux formation in the horse.

ALTHOUGH the erythrocyte sedimentation rate (ESR) has been well documented in the horse (Archer 1959, Schalm and others 1975, Jeffcott 1977, Hammerl and Kraft 1983) the mechanisms responsible for an increase or decrease in the rate of settling of the erythrocytes are not fully understood (Bottiger 1967, Osbaldiston 1971, Jain 1986). According to Osbaldiston (1971) the erythrocytes of the horse settle very quickly because of extensive rouleaux formation which occurs even in apparently healthy animals with normal plasma protein profiles. This ability of the erythrocytes to form rouleaux (reversible aggregation) depends upon a flexible erythrocyte membrane and the presence of adequate concentrations in the plasma of aggregating agents or other macromolecules. According to Amin and Sirs (1985) the very high flexibility of equine erythrocytes is consistent with the enhanced rouleaux

formation seen in horse blood. In man, it has been shown that the most important plasma factor relating to increased rouleaux formation is fibrinogen (van Haeringen and Glasius 1970). Bottiger (1967) suggested that in the horse, fibrinogen may not induce red cell aggregation to any great extent, but that plasma β globulins could be a more important factor. This was not the view of Osbaldiston (1971) who considered that the biophysical properties of the equine red cell, rather than any specific plasma protein fractions, caused the rapid settling times. Whatever the mechanisms involved in the hyperaggregation may be, the haematocrit has been shown to influence sedimentation rates, even over the physiological range (Torten and Schalm 1962). Recently a breed difference in the erythrocyte sedimentation rate has been observed. Hot blooded horses, including the English thoroughbred, have a relatively slow settling rate and consequently a low sedimentation rate compared with cold blooded horses which have a rapid settling rate (Hammerl and Kraft 1983). Whether these breed differences in the ESR are caused by different red cell characteristics, or, variations in plasma aggregating agents or haematocrit, is not yet known.

The physiological aggregation of erythrocytes (rouleaux and rouleaux networks) is known to be responsible for the non-Newtonian behaviour of blood. When hyperaggregation of the erythrocytes occurs the normal functioning of the microcirculation is affected, and there is a decrease in the rate of supply of oxygen to the tissues. It is important, therefore, that the plasma factors which influence rouleaux formation and red cell clumping (reversible aggregation and irreversible agglutination) are clearly identified. By measuring the ESR (with and without haematocrit standardisation), plasma viscosity and the concentrations of plasma proteins, it has been possible to assess the influence of fibrinogen, total serum globulins and the haematocrit on the sedimentation rates in a population of young thoroughbred racehorses.

Materials and methods

A total of 138 blood samples were collected from 123 un-

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excited, resting thoroughbred racehorses aged two to four years which were at various stages of training and fitness. Of these, 105 were haematologically and biochemically normal horses and 18 had abnormal plasma proteins as a result of minor inflammatory disorders, including upper respiratory tract infections. Fifteen horses were bled twice, once when healthy and again when sick. For the purpose of this investigation exact clinical diagnoses were not considered. The blood sampling procedure and laboratory techniques have been described elsewhere (Allen and Powell 1983, Allen and Kold 1988) with the exception of the following. The first 5 ml of blood withdrawn was mixed with potassium ethylene diamine tetra acetic acid (K_2 EDTA) (1.5 mg/ml) and used for the resting blood count as recommended by Rose and Allen (1985). The 60 minute ESR, based upon the Westergren technique, was performed using commercially available Sedioplast tubes (Lea Scientific) at room temperature (20° to 22°C) on blood containing potassium EDTA which was mixed immediately before testing with 31.3 g/litre trisodium citrate (one part of citrate to four parts of blood) as follows. Each horse's blood was adjusted by the removal or addition of native plasma to give five samples with different haematocrits in the range 0.28 to 0.48 on which the ESR and haematocrit were measured. One of these samples was the resting blood count sample which was not reconstituted. A line of best fit of ESR against haematocrit was then drawn and two ESR values, one corresponding to a standard haematocrit of 0.40 (referred to as the haematocrit standardised ESR) and the other corresponding to the horse's original resting haematocrit (referred to as the ESR) were obtained by interpolation. The haematocrit values used for this graph were measured in the potassium EDTA blood samples before dilution with citrate. All blood samples were collected into plastic tubes between 08.45 and 09.00 and the ESR measured within one hour of sampling.

Results

Table 1 shows the correlation coefficients between fibrinogen, total serum globulin, albumin, plasma viscosity and leucocyte counts, and the ESR and the haematocrit standardised ESR. The linear relationship between fibrinogen and haematocrit standardised ESR (Fig 1) was better than the relationship between fibrinogen and ESR (Fig 2).

There was a highly significant negative correlation between haematocrit and ESR values ($r = -0.640$, $P < 0.001$). This curvilinear relationship is shown in Fig 3. The haematocrit showed a strong positive correlation with albumin ($r = 0.504$, $P < 0.001$) but no significant correlation with fibrinogen ($r = 0.118$) serum total globulins ($r = 0.100$) or plasma viscosity ($r = 0.036$).

To study the characteristics of horses with abnormally high fibrinogen levels (>4.5 g/litre) the horses were divided into two groups, one with fibrinogen levels of 3.0 to 3.5 g/litre and the other with fibrinogen levels of 4.5 to 6.5 g/litre. The results are shown in Table 2.

Discussion

The haematocrit ($r = -0.64$) fibrinogen ($r = 0.61$) plasma viscosity ($r = 0.59$) and serum total globulins ($r = 0.38$) were all significantly correlated with ESR in this population of young thoroughbred racehorses. When the ESR was standardised to a common haematocrit, to remove the influence of this factor, the correlation coefficients for the plasma factors were all higher but remained in the same order as for the non-standardised ESR. These results do not entirely agree with earlier studies which suggested that increases in fibrinogen and other acute phase proteins do not greatly influence the ESR (Bottiger 1967, Archer and Allen 1970, Osbaldiston 1971, Jain 1986). The improved method of measuring ESR used here, together with a rigid blood sampling procedure and

TABLE 1: Correlation coefficients (r) between measurements of ESR and haematocrit standardised ESR, and plasma protein concentrations plasma viscosity and leucocyte counts

	ESR		Haematocrit Standardised ESR	
	r	P	r	P
Fibrinogen	0.614	<0.001	0.754	<0.001
Total serum globulin	0.382	<0.001	0.424	<0.001
Albumin	-0.349	<0.001	-0.138	NS
Plasma viscosity	0.588	<0.001	0.735	<0.001
Total leucocyte count	0.025	NS	0.064	NS
Absolute neutrophil count	0.027	NS	0.182	NS

NS Not significant

careful selection of a single breed of horse, may account for the different findings.

The results of the linear regression analysis which showed that haematocrit and fibrinogen were independent influences on red cell aggregation in this population of horses, agree well with the observations found in man by Bain (1983). Because haematocrit and ESR are negatively correlated any conditions which increase the haematocrit, such as excitement, exercise or dehydration, will tend to decrease the ESR, whereas conditions which reduce the haematocrit, for example anaemia, will tend to increase the ESR. Furthermore, the increase in haematocrit values observed after training programmes (Allen and Snow 1986) will result in a corresponding decrease in ESR with increasing fitness, provided that other variables remain constant. It is likely that physiological variability in

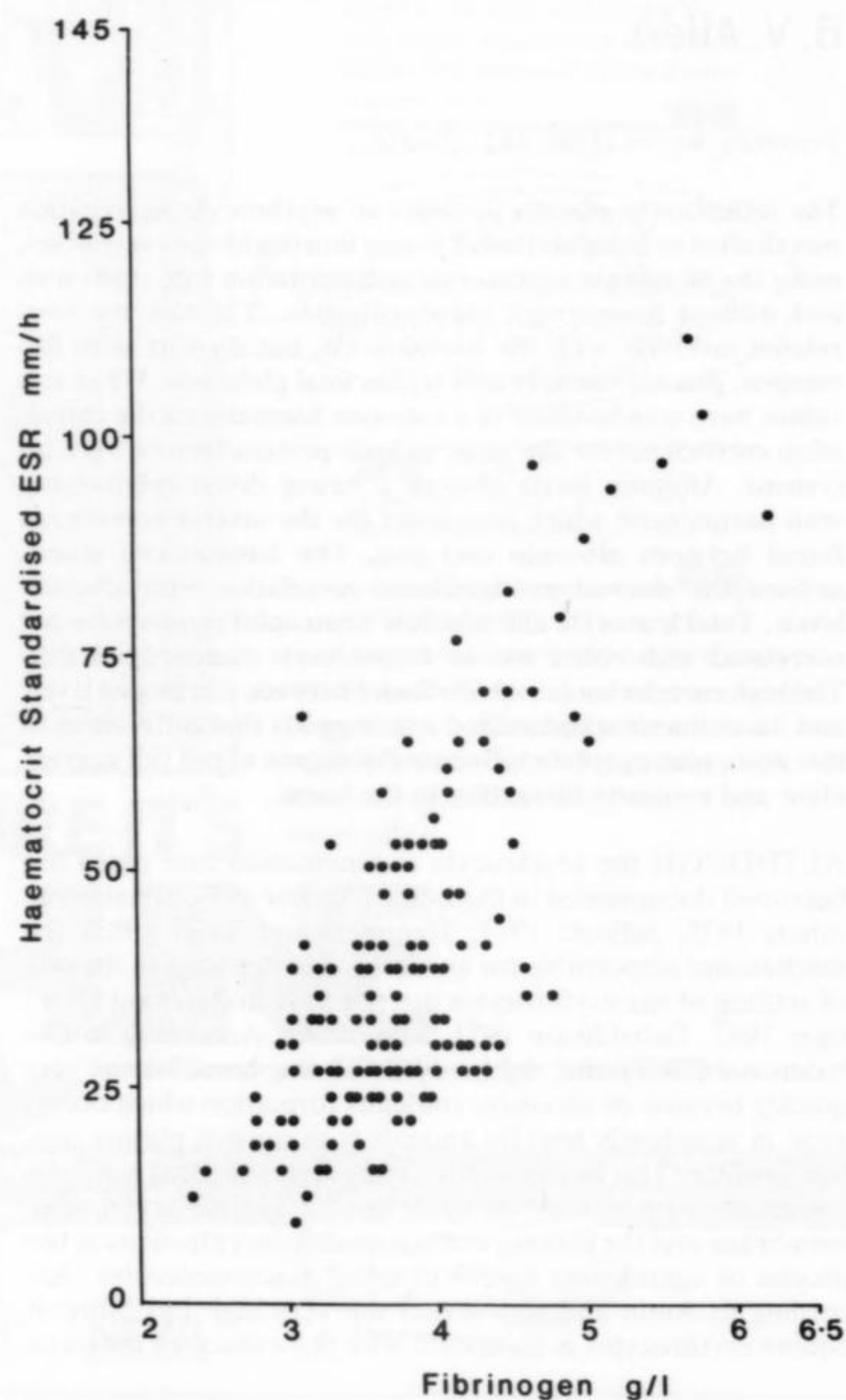


FIG 1: Relationship between fibrinogen and haematocrit standardised ESR in 123 thoroughbred horses, $y = -49.1 + 23.3x$, $n=138$

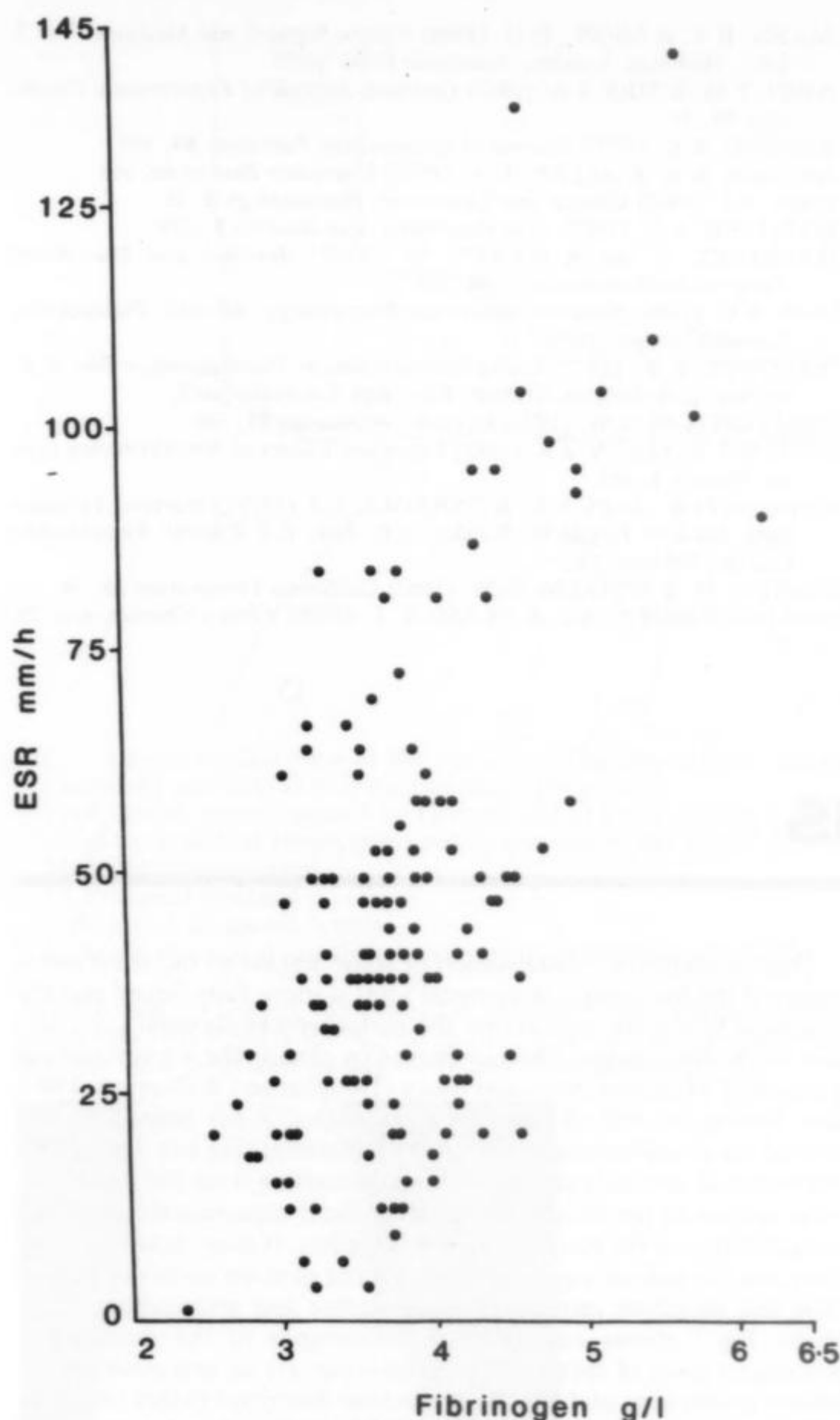


FIG 2: Relationship between fibrinogen and ESR in 123 thoroughbred horses. $y = -48.9 + 24.7x$. $n=138$

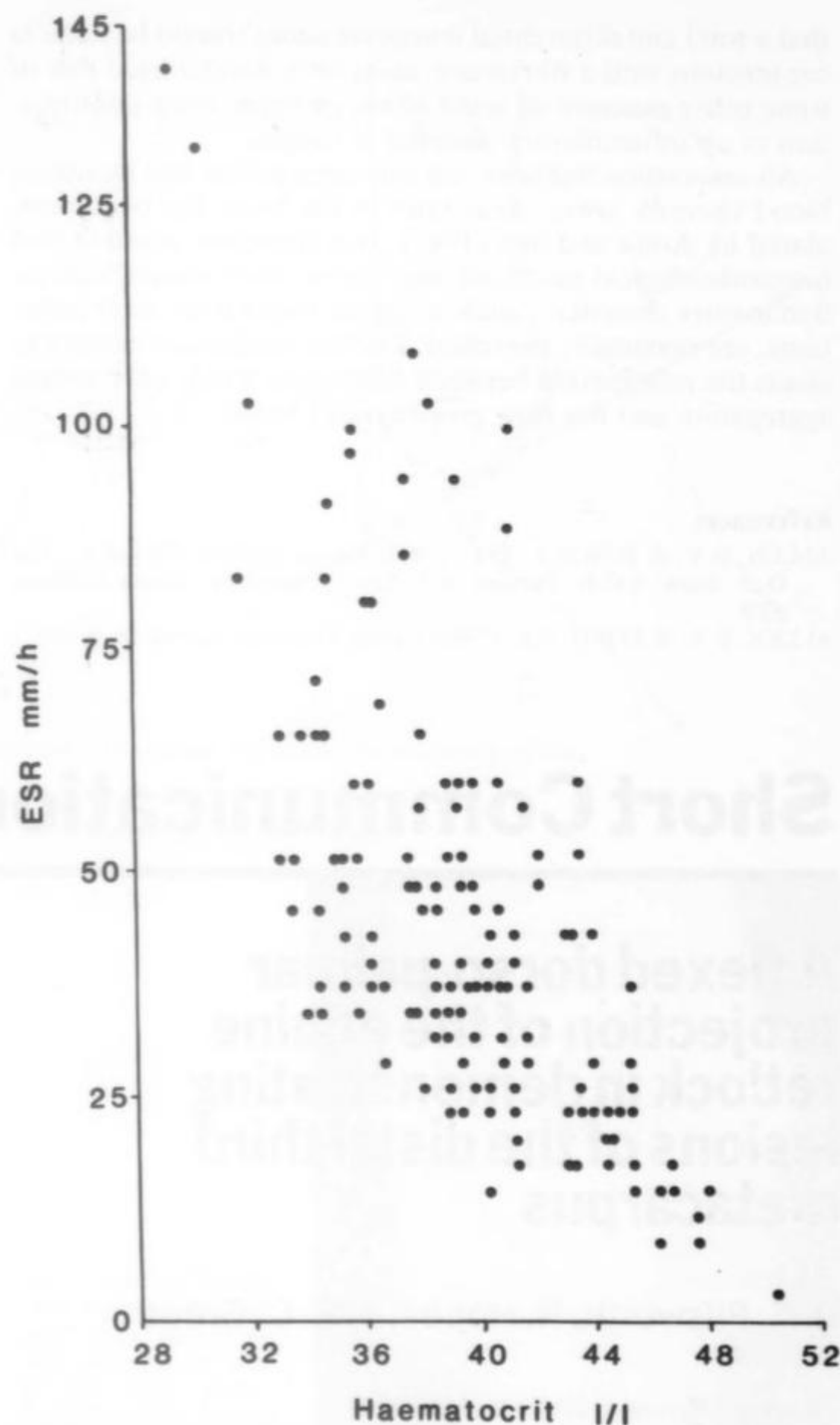


FIG 3: Relationship between haematocrit and ESR in 123 thoroughbred horses. $y = 192.7 - 3.89x$. $n=138$

the haematocrit, and to a lesser extent fibrinogen concentrations, is responsible for much of the differences in the ESR between horses.

However, in man, several factors other than haematocrit and fibrinogen levels can influence the ESR. These include the concentration of various immunoglobulins, complexes between fibrinogen and immunoglobulins and increases in specific globulin fractions, as well as differences in the physical properties of the red cell such as size, shape and flexibility. According to Amin and Sirs (1985) there is a factor in horse plasma which increases the flexibility of the red cells and therefore their rate of aggregation and rouleaux formation. Variations in this factor from horse to horse may also cause differences in the ESR even when the levels of fibrinogen and other globulins remain within normal limits. The inverse correlation between albumin and ESR was probably due to the correlation between the ESR and the haematocrit, which was also directly related to albumin. Furthermore, when the ESR was standardised to remove the influence of variations in haematocrit, the relationship between ESR and albumin disappeared. The direct relationship observed between albumin levels and haematocrit is not easy to explain. However, albumin and several important haematinic factors are either synthesised or stored in the liver, and it is therefore possible that this organ may have been responsible, either directly or indirectly, for the relationship observed.

The haematocrit adjusted ESR is influenced mainly by changes in the concentrations of fibrinogen and other plasma

globulins, which should make it a fairly specific test for inflammation. The lack of a correlation between total leucocyte and absolute neutrophil counts and ESR, whether haematocrit standardised or not, was therefore surprising because the white cells are often useful indicators of inflammatory disease, and the ESR has been shown in this study to be related to fibrinogen and total serum globulins, both of which increase in inflammatory conditions. Total leucocyte and absolute neutrophil counts were only slightly higher in the group of horses with high fibrinogen levels than in the horses with normal fibrinogen values, even though the ESR, total serum globulin and plasma viscosity were all significantly higher in the horses with high fibrinogen levels (Table 2). These findings suggest

TABLE 2: Comparison of plasma proteins, ESR, plasma viscosity and leucocyte counts in 30 horses with normal fibrinogen levels and 17 horses with high fibrinogen levels

	Fibrinogen (g/litre)		P*
	(3.0-3.5)	(4.5-6.5)	
Total serum globulin (g/litre)	29.3±3.3	32.0±3.8	<0.02
Albumin (g/litre)	28.2±2.4	27.5±1.4	NS
Plasma viscosity (cP)	1.44±0.05	1.57±0.05	<0.001
ESR (mm/hour)	37±16	91±31	<0.001
Standardised ESR (mm/hour)	26±8	71±24	<0.001
Total leucocyte count (10 ⁹ /litre)	8.0±1.0	8.7±1.3	NS
Absolute neutrophil count (10 ⁹ /litre)	3.6±0.6	4.0±0.8	NS

* Unpaired t test, NS Not significant

that a total and differential leucocyte count should be done in conjunction with a fibrinogen assay or a standardised ESR or some other measure of acute phase proteins when confirmation of an inflammatory disorder is sought.

An association between red cell aggregation and increased blood viscosity lower shear rates in the horse has been postulated by Amin and Sirs (1985). It is therefore possible that haemorheological problems may occur when horses with inflammatory disorders, such as upper respiratory tract infections, are maximally exercised. Further studies are needed to assess the relationship between fibrinogen levels, erythrocyte aggregation and the flow properties of blood.

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Short Communications

A flexed dorso-palmar projection of the equine fetlock in demonstrating lesions of the distal third metacarpus

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LAMENESS originating from the metacarpo-phalangeal joint is a common presenting condition in the thoroughbred in training, comprising 14 per cent of all lameness diagnosed in one survey (Rossdale and others 1985). Clear radiological signs of disease in this joint are often strong prognostic indicators, as the joint may have suffered advanced degenerative changes in articular cartilage before radiographic abnormalities develop. Most clinics in this country employ a standard set of projections to investigate the metacarpo-phalangeal joint which comprises latero-medial, flexed latero-medial, proximodorsal-palmarodistal, dorsolateral-palmaromedial oblique and dorsomedial-palmarolateral oblique views. Recently, the present authors have been made aware of a novel projection which has dramatically highlighted lesions of the palmar aspect of the distal third metacarpus which were invisible on the standard proximodorsal-palmarodistal projection (G. Ferraro, personal communication). It appears that this view does not seem to have gained widespread use in the UK, but is now frequently employed in the USA. However, in a recent authoritative American review of radiography of the metacarpo-phalangeal joint, the projection is not listed (Stashak 1987). The purpose of this short communication is to bring this information to the notice of those clinicians in the UK who are not yet familiar with it.

The flexed dorso-palmar projection was first described by workers in Davis, California attempting to better visualise the lesions of palmar metacarpal disease (Hornof and O'Brien 1980). In their study the fetlock was flexed by placing the sole of the foot on a large elevated wooden block and an X-ray beam parallel to the horizontal was angled at approximately 125° to the long axis of the third metacarpal bone. The authors suggested that one quarter to one third of the proximal sesamoid bones should overlap the metacarpo-phalangeal joint on the final radiograph (Fig 1b).

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There is improved visualisation of some lesions of the third metacarpus if the metacarpo-phalangeal joint is more fully flexed and the sesamoid bones are well above the metacarpo-phalangeal joint surface on the final image. This is achieved by placing the toe vertically in a standard 'Hickman' navicular block (Douglas and Williamson 1980) thus flexing the fetlock, and using an angled X-ray beam with the central ray at approximately 35° to the horizontal (Fig 1c). This allows resolution of vertical radiolucent lines emanating from the distal articular surface of the third metacarpus without superimposition of the vertical edges of the proximal sesamoid bones. It does, however, superimpose the palmar aspect of the proximal phalanx on to the field of view and therefore introduces some scatter and attenuation of the beam. Fig 2 shows examples of radiographs of the metacarpo-phalangeal joint of three horses employing: (i) an orthodox dorso-palmar projection; and (ii) the projection described in this study. In each case, lesions were only identifiable in the flexed non-weight bearing dorso-palmar projection.

Since these cases have come to light, increasing use has been made of this projection in routine radiography of the metacarpo-phalangeal joint in this clinic with similarly encouraging results. Other studies have previously shown the value of the projection developed by Hornof and O'Brien (1980) in highlighting comminution of the distal extremity of condylar fractures (Rik and others 1983) and in detecting the presence of palmar metacarpal disease (Hornof and O'Brien 1980). Although the authors in no way make any claim for developing this projection, it is recommended to colleagues working in this area who are not already utilising it.

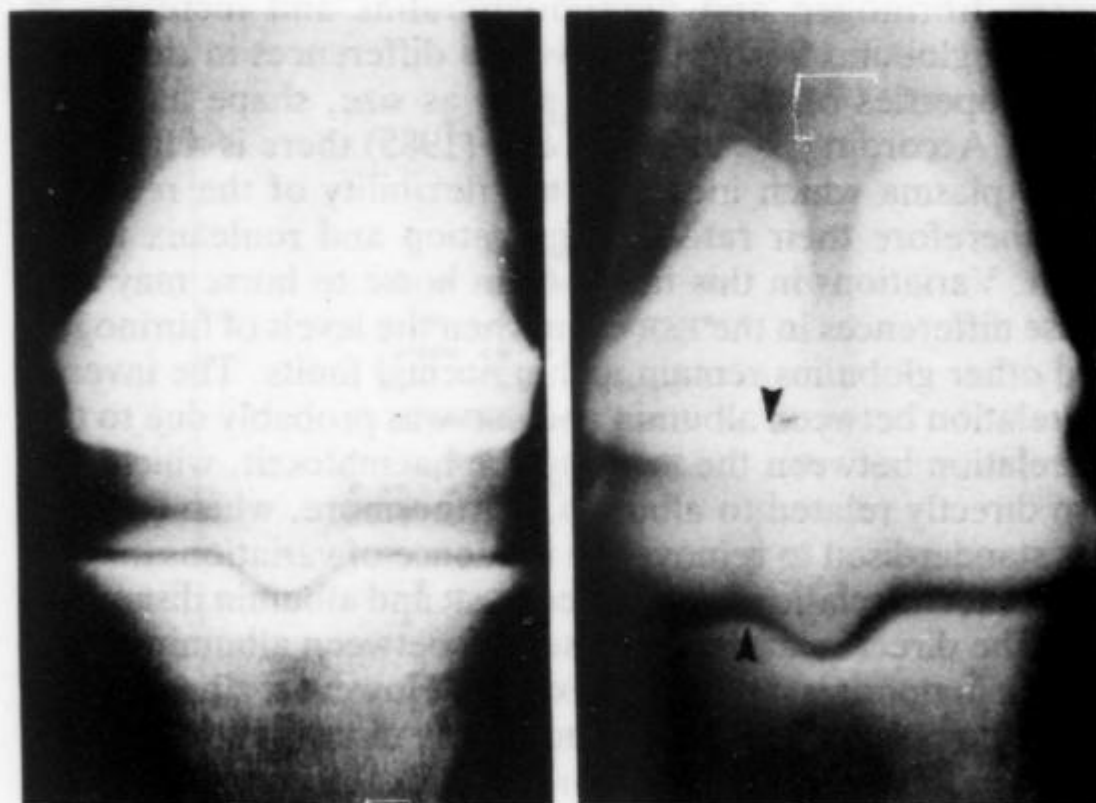


FIG 2: (a) (i) and (ii) Incomplete condylar fracture (arrows) which presented as acute onset lameness localised to the metacarpo-phalangeal joint by regional analgesia. No radiographic pathology shown using orthodox views